

DIABETES – glucose control

What diabetes is DEFINITION

Diabetes — a syndrome which has hyperglycaemia as its hallmark, due to an **absolute** lack of insulin (Type 1, formerly insulin dependent diabetes, **IDDM**) or a **relative** lack of insulin (Type 2, formerly non-insulin dependent diabetes, **NIDDM**).

Normal glucose 3.5–7 mmol/ml

Type 1 absolute lack

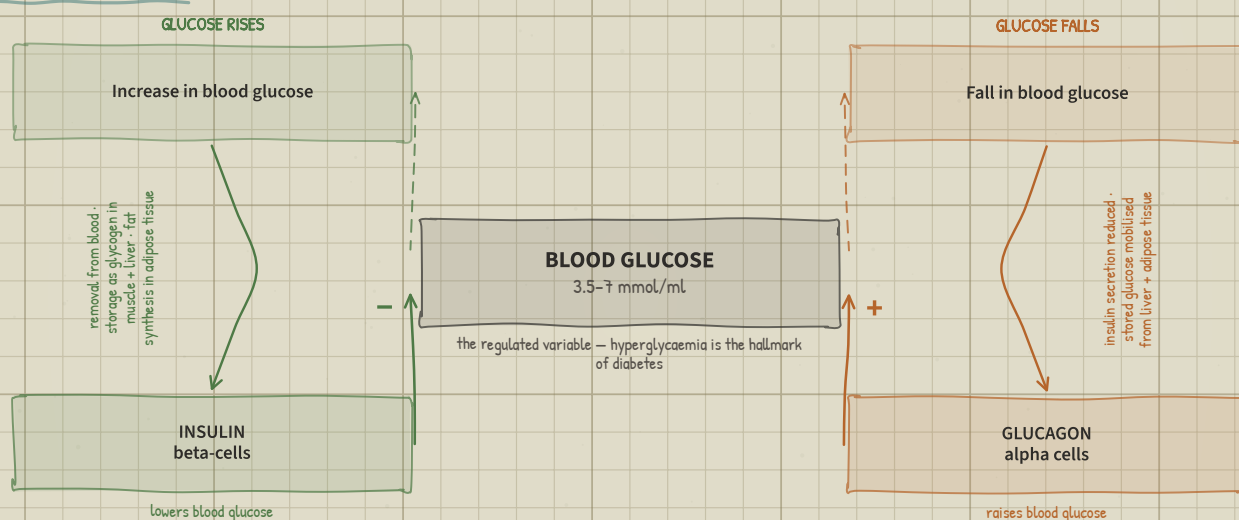
Type 2 relative lack

Normal control

Blood glucose is held at 3.5–7 mmol/ml by two opposing pancreatic hormones.

- **Insulin** — produced by the **beta-cells** of the pancreas.
- **Glucagon** — produced by the **alpha cells** of the pancreas.
- A rise in glucose stimulates insulin, which removes glucose from blood and stores it as **glycogen** in muscle and liver.
- Insulin also stimulates **synthesis of fat** from adipose tissue, stored as triglyceride.
- A fall in glucose reduces insulin and stimulates glucagon, which **mobilises stored glucose** from liver and adipose tissue.

Glucose homeostasis



Diabetes in the UK

1.4M

have a diagnosis of diabetes

~3%

of the population

1M+

more thought to be undiagnosed

Diabetes becomes more common with age: Type 2 is present in 5–10% of over-65s.

Primary vs secondary

The **majority** of cases are **primary** diabetes.

Group	Cause
Primary	The majority of all cases of diabetes
Secondary	Pancreatic carcinoma
	Liver disease, particularly cirrhosis
	Gestational diabetes mellitus

Who gets it

- **Men** are one and a half times as likely to be diagnosed as women — *except* women who are overweight.
- **IDDM** is very rare under **9 months** of age; peak incidence of onset is **5–15 years**.
- It is the **second commonest chronic illness of childhood**.
- About **30%** of IDDM cases present in patients **older than 30 years**.

Late-onset Type 1 CLINICAL NOTE

In older patients symptoms can be **less dramatic** and may initially be thought to be NIDDM. These patients are usually **non-obese**, have a history of **weight loss**, and respond **poorly to oral hypoglycaemic agents**.

Exam point

- Insulin is the **only** hormone that lowers blood glucose; glucagon, and the loss of insulin, both raise it.
- Type 1 = **absolute** lack. Type 2 = **relative** lack. This single distinction drives every difference in presentation and treatment.
- **30%** of IDDM presents over the age of 30 — a non-obese adult with weight loss who fails oral agents is Type 1 until proven otherwise.

TYPE 1 vs TYPE 2 – causes & presentation

Cause of Type 1 (IDDM) MECHANISM

An **autoimmune disorder** characterised by pancreatic beta-cell destruction. Dietary factors may also be involved: absorption of **cow's milk proteins** through the immature gut (i.e. at 3 months of age) may set off an immune response leading to destruction of the beta-cells.

Cause of Type 2 (NIDDM) MECHANISM

Described as a lifestyle disease, developing in people who are **genetically predisposed**, where environmental factors — nutrition (particularly obesity) and exercise — modulate expression of the diabetic phenotype.

- A **family history of Type 2** remains the *strongest single risk factor*.

Three abnormalities in Type 2

TYPE 2 DIABETES three main abnormalities

Peripheral insulin resistance

- Defect of the insulin receptor mechanism
- Decreased NUMBER of receptors, or
- Decreased SENSITIVITY of receptors
- Glucose uptake falls → hyperglycaemia

Increased hepatic glucose production

- Liver continues to release glucose
- Adds to the circulating load

Reduced pancreatic insulin secretion

- Relative — not absolute — lack of insulin
- Distinguishes Type 2 from Type 1

A receptor defect — fewer receptors, or reduced receptor sensitivity — reduces glucose uptake and causes hyperglycaemia.

Defining features

TYPE 1

- Sudden onset
- Severe symptoms, including ketoacidosis — coma in some
- Recent weight loss
- Onset more acute; many are admitted to hospital

VS

TYPE 2

- Gradual onset
- May be no symptoms
- Often no weight loss
- Usually obese
- Not ketotic
- Slower onset of symptoms

Signs and symptoms

- Increased urine output (**polyuria**) and excessive thirst (**polydipsia**).
- **Weight loss and weakness** due to tissue breakdown.
- **Blurred vision**.

The receptor defect MECHANISM

A defect of the **insulin receptor mechanism** — either a *decreased number* of receptors or *decreased sensitivity* of the receptors — reduces glucose uptake, causing hyperglycaemia.

Fewer receptors ↓ uptake

Less sensitive ↓ uptake

Net effect hyperglycaemia

Why Type 2 is missed COMMON MISTAKE

Patients may attribute vague symptoms such as **dry mouth** and **poor concentration** to increasing age, not to diabetes. Only as glucose values rise do polyuria, thirst or weight loss become apparent and cause patients to seek help.

Where the symptoms come from

- **Polyuria** — glucose spills into urine and drags water with it.
- **Polydipsia** — thirst follows the urinary water loss.
- **Weight loss and weakness** — tissue breakdown, because glucose cannot be used as fuel.
- **Blurred vision** — an early and often-ignored complaint.

Exam point

- The **strongest single risk factor** for Type 2 is a *family history of Type 2* — not obesity, though obesity modulates how the phenotype is expressed.
- Type 1 is **autoimmune beta-cell destruction**; Type 2 is a *triad* — peripheral insulin resistance, increased hepatic glucose production, and reduced pancreatic insulin secretion.
- **Ketosis separates the two**: Type 1 presents with ketoacidosis, Type 2 is not ketotic.

DIAGNOSIS & acute complications

How the diagnosis is made

ROUTE 1 — CLASSIC SYMPTOMS PRESENT
plus any ONE qualifying plasma glucose result

→ Symptoms: polyuria, polydipsia, weight loss, blurred vision

ROUTE 2 — ASYMPTOMATIC PATIENT

Two venous samples from any of the categories below, on TWO DIFFERENT DAYS

→ Without symptoms a single abnormal result is not enough

Diagnosis is usually made by one of these two routes.

Qualifying results

Test	Threshold	Condition
Fasting plasma glucose	> 7.0 mmol/mol	Fasted sample
Random plasma glucose	> 11.1 mmol/mol	Any time
Oral glucose tolerance test (OGTT)	> 11.1 mmol/mol	2 hours after 75 g anhydrous glucose

Diabetic ketoacidosis DEFINITION

DKA — an **acute, major, life-threatening** complication of diabetes, characterised by hyperglycaemia, ketoacidosis and ketonuria.

Initial presentation in up to 16% of Type 1

Mortality 2-15%

What precipitates DKA

It is a **fairly common but serious** event, precipitated by:

- **Infection** — e.g. a urinary tract infection
- **Acute illness**
- **Trauma**
- **Missed insulin doses**

Mechanism of DKA

Absolute or relative **INSULIN DEFICIENCY**

→ The initiating defect

Glucose cannot enter cells for utilisation as metabolic fuel

→ Glucose accumulates in blood → hyperglycaemia

Liver rapidly breaks down **FAT** into **KETONES** to employ as a fuel source

→ The body's emergency fuel switch

Overproduction of ketones — they accumulate in blood and urine

→ Ketonuria

BLOOD TURNS ACIDIC — ketoacidosis

→ Breath smells sweet, of acetone

Lack of insulin means glucose cannot be utilised as an energy source — so the liver burns fat instead.

Symptoms of DKA

- Thirst, polyuria, dehydration
- Vomiting, hyperventilation, weakness
- **Ketosis** — breath smells sweet, of acetone
- Tachycardia, hypotension
- Drowsiness, confusion, **coma**

Non-ketotic hyperglycaemia MECHANISM

Occurs in patients with **Type 2 diabetes**. There is **no** production of ketones because there *is* sufficient insulin present to prevent excessive lipolysis — but not enough for complete utilisation of glucose, hence hyperglycaemia.

- As there are no ketones, there is **no vomiting** and **no metabolic acidosis**.

Exam point

- The diagnostic thresholds are **fasting > 7.0 mmol/mol** and **random or 2-hour OGTT > 11.1 mmol/mol**. With symptoms, one result is enough; without symptoms, you need **two samples on two different days**.
- DKA vs non-ketotic hyperglycaemia is a question about *how much insulin is left*: enough to suppress lipolysis (no ketones, Type 2) or not (ketones, Type 1).

CHRONIC complications

The complications DEFINITION

They include **hypoglycaemia** and problems relating to hyperglycaemia — e.g. *cystitis*, *skin infections* and *gingivitis* — but the most important are the vascular complications (nephropathy and cardiovascular disease).

- These take an average of **15–20 years** to develop in **Type 1**.
- They may be encountered **more quickly** in Type 2 — indeed **may be present at diagnosis**.

Hypoglycaemia — causes

The **most common** complication in diabetics. In most cases the patient or carer should be able to initiate treatment.

- **Insufficient food intake** — skipping meals, vomiting, diarrhoea
- **Excessive exercise**
- **Hypoglycaemic drugs**
- **Too much insulin**

Hypoglycaemia — symptoms

- Nausea, hunger
- Tachycardia, **paraesthesia in the fingers or lips**, excessive sweating, muscle tremor
- Ataxia, blurred vision, headache, confusion, agitation, **coma and convulsions**

Treating hypoglycaemia

ORAL GLUCOSE
initial treatment

→ Patient or carer can usually start treatment themselves



INTRAVENOUS GLUCOSE — 50%, 20–50 ml
in severe cases

→ If the patient cannot swallow safely



GLUCAGON 1 mg by DEEP IM INJECTION into the buttock
if IV not available or cannot be given

→ Effect within minutes — it increases plasma glucose by mobilising glycogen stored in the liver

Rapid treatment is essential to avoid permanent cerebral damage.

Retinopathy CLINICAL NOTE

Diabetic retinopathy is the **commonest** cause of blindness. In Type 1 it starts after **10 or more years**; in Type 2 it may be **present at diagnosis**.

Have retinopathy ~30% of all diabetics

Laser, if detected early prevents 70% going blind

Eye checks yearly

Neuropathy CLINICAL NOTE

Damage to **sensory, motor and autonomic** nerves, commonly occurring as a result of chronic hyperglycaemia.

- Impaired nerve function found in **60%** of diabetics; causes problems for **about a third** of them.
- Commonest manifestation: **reduced sensation in the feet** — an important factor in foot ulceration.
- Peripheral neuropathy also impairs **sweating**, leading to dry, cracked skin that may fail to heal.
- If ulceration is due to **peripheral vascular disease**, amputation is often needed.

Nephropathy CLINICAL NOTE

Diabetic nephropathy affects:

- About **one in every three** individuals with **Type 1** diabetes.
- About **one in every 20** with **Type 2**.
- All people with diabetes should have urine tested **at least annually** for **proteinuria** and, if negative, for **microalbuminuria**.

Key takeaway — the feet

It is essential that each diabetic patient is **individually educated about foot care**, as most diabetic foot problems are entirely preventable with good foot care and regular screening.

TREATMENT – insulin

Aim of treatment DEFINITION

To achieve the best possible control of plasma glucose without inducing hypoglycaemia, in order to prevent both the **micro-** and **macrovascular** complications of diabetes.

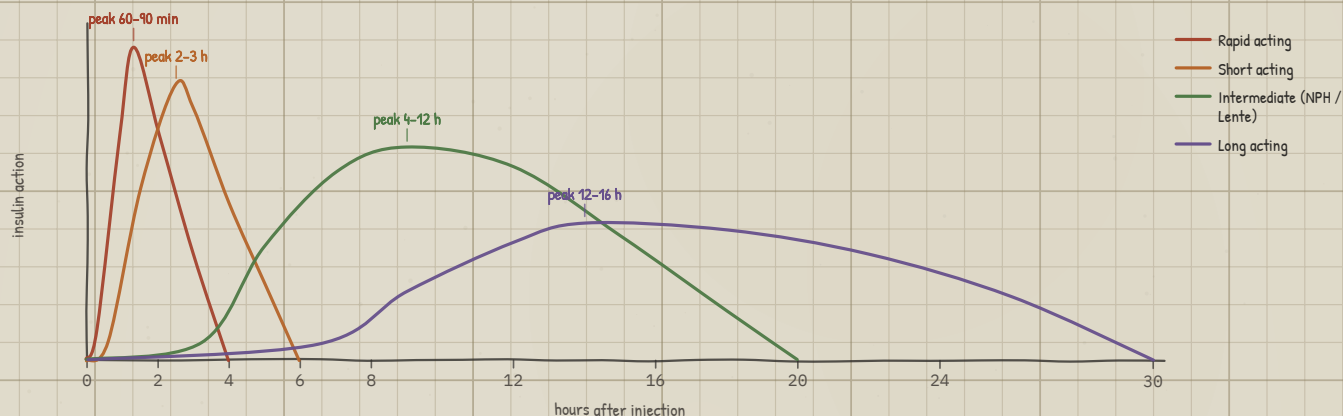
Diet the cornerstone – both types

Type 1 match energy + insulin intake

Type 2 many controlled by diet alone

Exercise lowers glucose + increases insulin action

Insulin action profiles



Schematic curves drawn from the onset / peak / duration figures given in the source – the shapes show why the classes are combined.

Insulin therapy – the four classes

Class	Onset	Peak	Duration	How it is used
Rapid acting	15 min	60-90 min	2-4 h	Before a meal, to cover the glucose rise from eating. Used with a longer-acting insulin.
Short acting	30-60 min	2-3 h	4-6 h	About 30 min before a meal, to cover the glucose rise from eating. Used with a longer-acting insulin.
Intermediate (NPH / Lente)	3-4 h	4-12 h	16-20 h	Covers glucose rises when rapid-acting insulins stop working. Often combined with rapid or short acting; usually twice a day .
Long acting	6-8 h	12-16 h	20-30 h	Lowers glucose when rapid-acting insulins stop working. Combined when needed; taken once or twice a day .

Starting dose WORKED EXAMPLE

GIVEN A patient of 70 kg, newly started on insulin

FORMULA dose = 0.5 units / kg / day

SUBSTITUTION 0.5×70

35 units per day

The dose is then adjusted according to the patient's blood glucose results.

Where insulin comes from

Insulin is categorised by **how it is derived** and **how quickly it acts** – from rapid to long acting, from animal insulin through to analogue insulins.

- **Animal insulin** was the first type given to humans, derived from *cows* and *pigs*.
- Until the **1980s** it was the only treatment for insulin dependent diabetes.
- It has largely been replaced by **human insulin** and **human analogue insulin**.

Giving the injection PROCESS

- 1 Give **30-45 minutes before food** where possible.
- 2 Inject **subcutaneously** – front of the thighs, lower abdominal wall, upper arms.
- 3 **Rotate sites daily** to prevent fibrosis, which interferes with insulin absorption.

What speeds absorption WARNING

- **Hot baths and showers** increase the absorption rate.
- **Leg exercise** increases the absorption rate from the thigh.

Insulin devices

- Vial + syringe
- Penfill cartridge + injection device
- Flexpen (ready filled)

TYPE 2 – drug treatment

Components of treatment

Treatment varies from patient to patient; the aims are met by a **combination** of:

- Dietary adjustment
- Weight loss in the obese
- Adequate physical activity
- Oral antidiabetic drugs
- Patient education

Dietary advice

Unless a patient presents with severe symptoms, **diet and lifestyle modification alone** should be tried for at least **3 months**. Diet alone can control up to **30%** of patients with Type 2.

- Reduce energy intake if obese — to **1000–1500 kCal** per day.
- Eat **regular meals**; include a wide variety of foods.
- **Limit** intake of sugar and sweet foods.
- **Reduce** total fat intake.

Drug escalation

DIET AND LIFESTYLE MODIFICATION ALONE
at least 3 months, unless severe symptoms

→ Continue dietary recommendations even after drugs are started

STANDARD-RELEASE METFORMIN
the initial drug treatment for adults with Type 2

→ If metformin is contraindicated or not tolerated, go straight to dual therapy

ADD a second agent — DUAL THERAPY
sulphonylurea · DPP-4 inhibitor · GLP-1 agonist · SGLT inhibitor

→ Added when metformin has not continued to control HbA1c

TRIPLE THERAPY
if dual therapy has not succeeded

→ Below the person's individually agreed HbA1c threshold

INSULIN-BASED TREATMENT
if there is still no control

→ After 10 to 20 years, almost all patients with Type 2 will need insulin

Drugs in dual therapy are introduced stepwise, checking tolerability and effectiveness of each drug.

Metformin (biguanide) MECHANISM

Main effect: a reduction in hepatic gluconeogenesis. It also **reduces intestinal glucose absorption** and **enhances insulin sensitivity**.

- Treatment of choice in **obese** patients, but *equally effective in non-obese*.
- Nausea, vomiting, flatulence and diarrhoea in up to **20%** of patients.
- Reduce these by starting low — e.g. **500 mg** daily with or after food — and increasing gradually.

Switching sulphonylurea COMMON MISTAKE

It is unlikely that **if one sulphonylurea fails** to control hyperglycaemia, another will do so for any length of time. Second generation agents are more potent, but there is *no evidence of clinically important differences in efficacy* within the group.

Emphasise this CLINICAL NOTE

It is important to emphasise to patients the need to **continue with dietary recommendations even if they are started on oral medication**.

Sulphonylureas MECHANISM

Main effect: **increasing the sensitivity of the pancreas to glucose**, thus increasing insulin output. Secondary effects: increased muscular uptake and reduced hepatic synthesis of glucose.

- **Primary failure** — **25–30%** of patients may not respond.
- **Secondary failure** — a further **5–10%** lose responsiveness *annually*.
- Second generation agents (glimepiride) are more potent, but there is **no evidence of clinically important differences in efficacy** within the group.
- Take **30 minutes before meals** for maximum benefit.

Sulphonylurea side-effects WARNING

Most side effects are **shared by all** sulphonylureas.

- **Hypoglycaemia** — common, especially in the *elderly* and those with *renal or hepatic impairment*.
- **Weight gain** — can limit their usefulness in obese diabetic patients.
- **GI disturbances** — usually only on initiation, and should gradually wear off.
- **Rash** in about **3%**, usually in the first **6–8 weeks**.
- **Raised LFTs**.

INCRETINS, SGLT & monitoring

The newer agents

Class	Example	Mechanism	Notes
GLP-1 agonist	Victoza	Stimulates glucose-dependent insulin release from the pancreatic islets	Injectable. Also slows gastric emptying, inhibits inappropriate post-meal glucagon release, reduces food intake
DPP-4 inhibitor (gliptins)	Sitagliptin, saxagliptin, linagliptin	Increases incretin levels (GLP-1 and GIP) → inhibits glucagon release → increases insulin secretion, decreases gastric emptying and blood glucose	FDA approved. Adverse effects usually mild and transient: nasopharyngitis, headache, dizziness, arthralgia, nausea, constipation
SGLT1 inhibitor	—	Delays and reduces glucose absorption in the small intestine	Improves post-meal control; useful where declining renal function makes SGLT2 inhibition less effective
SGLT2 inhibitor	Canagliflozin	Reduces renal tubular glucose reabsorption	Lowers glucose without stimulating insulin release. Favourable effects on blood pressure and weight

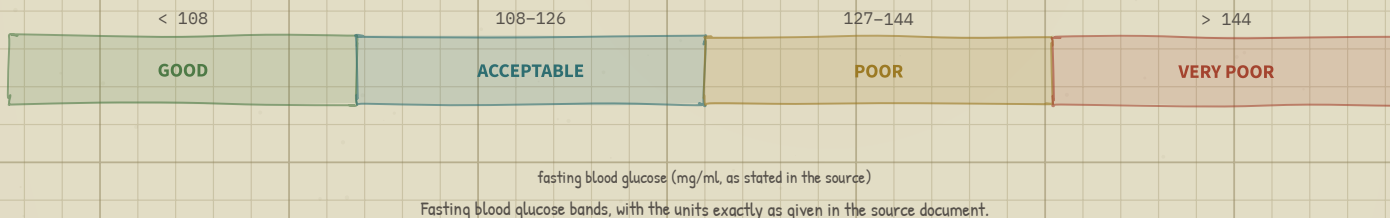
Note on glibenclamide CLINICAL NOTE

In patients **without renal or hepatic impairment**, glibenclamide may be used. They should be taken **30 minutes before meals** for maximum benefit.

The newer agents at a glance

- **GLP-1 agonists are injectable** — the others here are oral.
- DPP-4 inhibitors work **indirectly**, by raising incretin levels.
- SGLT2 inhibitors lower glucose **without stimulating insulin release**.
- SGLT2 also has favourable effects on **blood pressure and weight**.

Fasting blood glucose — what is good control?

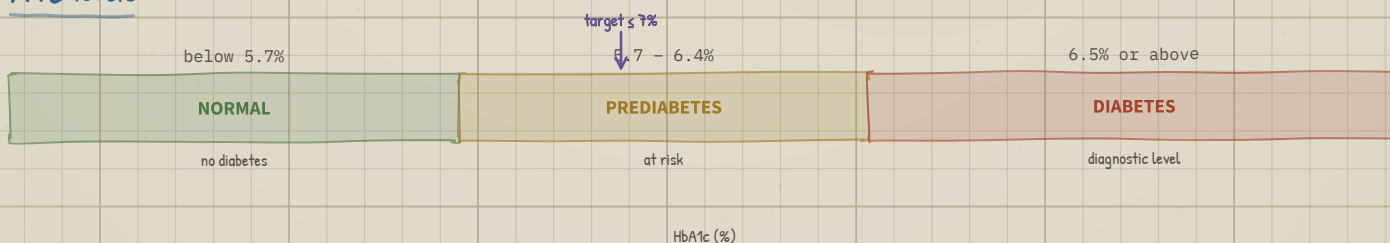


The A1c test (HbA1c) DEFINITION

A1c — a blood test giving a person's **average blood glucose over the past 3 months** — also called haemoglobin A1c, HbA1c. It is the **primary test used for diabetes management**, reported as a percentage: the higher the percentage, the higher the blood glucose levels have been.

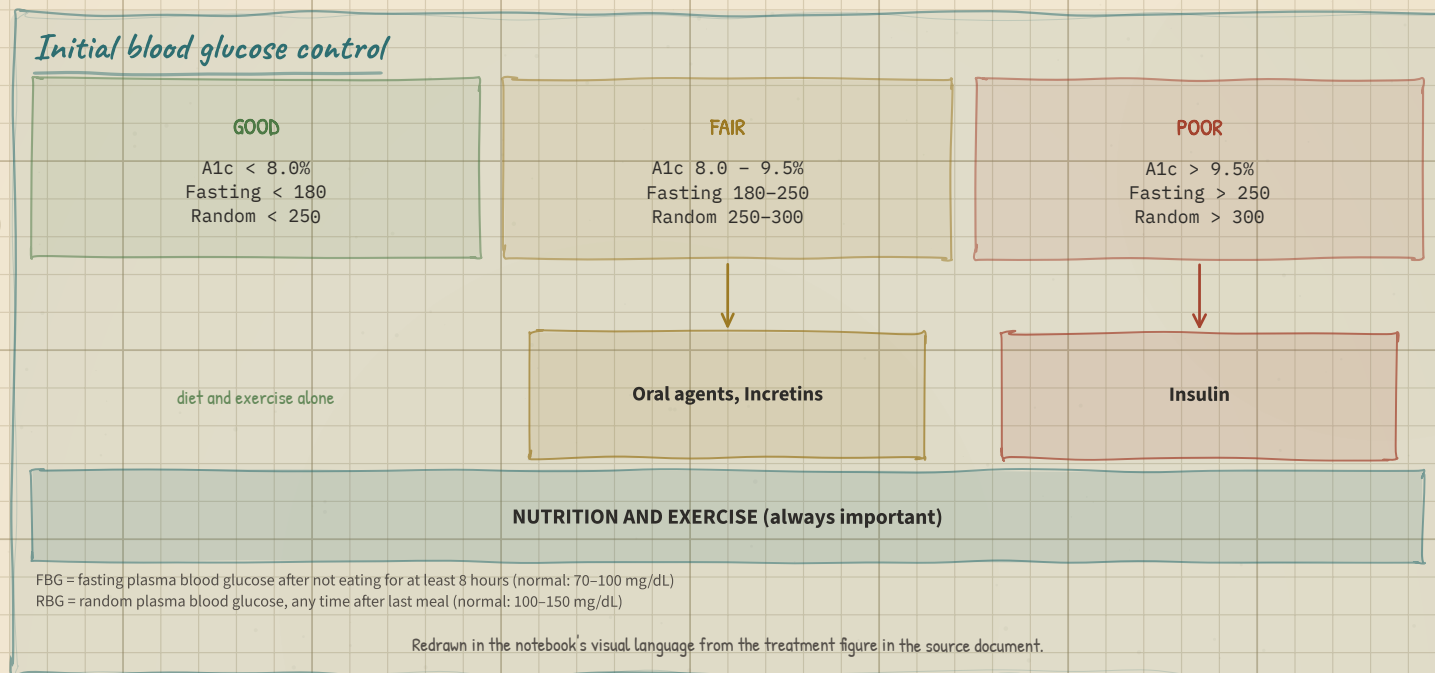
- RBCs are composed of **haemoglobin**. Glucose sticks to haemoglobin to make a **glycosylated haemoglobin molecule (HbA1c)**.
- The more glucose present in the blood, the more HbA1c will be present.
- RBCs are constantly forming and dying — they live for about **3 months**, so the test reflects the **average over that window**.

A1c levels



The treatment target for most people with diabetes is an A1c of ≤ 7%. Higher levels may need a more intensive medication plan.

THE CONTROL algorithm



When to start insulin in Type 2

Although treatment starts with oral medication and lifestyle changes, **most people with Type 2 eventually need insulin.**

- Significant hyperglycaemia in a **symptomatic** patient — “the need for insulin may be short-term”.
- Patients on **multiple diabetic medications** with uncontrolled diabetes.
- **Uncontrolled diabetes in pregnancy.**
- “After 10 to 20 years, almost all patients with type 2 diabetes will need insulin.”

Key takeaway

Control is not judged by a single number: the algorithm reads **A1c, fasting and random glucose together**, and every band sits on the same foundation of nutrition and exercise.

Exam point

- “The American Association of Clinical Endocrinologists recommends starting a person with type 2 diabetes on insulin if their A1C is above **9%** and they have symptoms.”
- Blood sugar control is **one of the most important parts** of Type 2 management.
- A1C reflects **3 months** — because that is how long a red cell lives.

A1C in pregnancy CLINICAL NOTE

- The A1C test may be used at the **first visit during pregnancy**, to see if women with risk factors had **undiagnosed diabetes before becoming pregnant**.
- **OGTT** is used to test for diabetes that develops *during* pregnancy.
- After delivery, women who had **gestational diabetes** should be tested for **persistent diabetes**.

Numbers to remember

≤ 7%

A1C treatment target for most patients

> 9%

A1C + symptoms → start insulin (AACE)

10–20 y

until almost all Type 2 need insulin

Summary – the Type 2 pathway

- **Diet and lifestyle alone** for at least 3 months — controls up to **30%** of patients.
- **Metformin** is the initial drug treatment; reduces hepatic gluconeogenesis.
- **Dual therapy** adds a sulfonylurea, DPP-4 inhibitor, GLP-1 agonist or SGLT inhibitor — stepwise, checking tolerability and effectiveness of each drug.
- **Triple therapy** if dual therapy does not succeed.
- **Insulin** if there is still no control — and after **10–20 years**, almost all Type 2 patients will need it.

THYROID – physiology & TFTs

How common

2%

of women have
hypothyroidism

0.13%

of men have
hypothyroidism

1–2%

UK prevalence of
hyperthyroidism

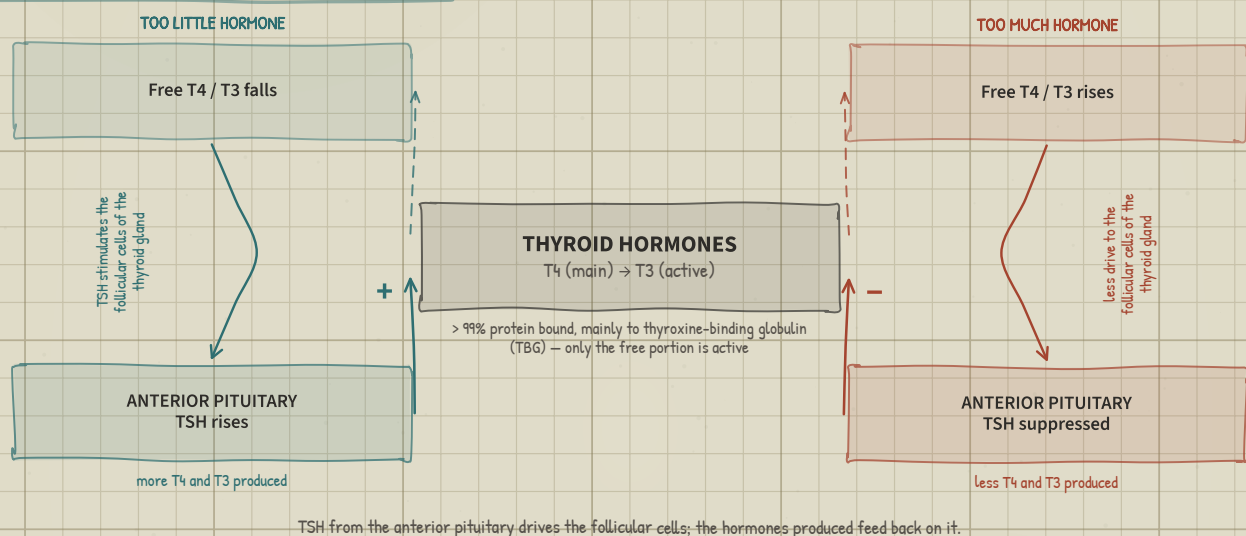
Most patients with hypothyroidism present between the ages of **30** and **60**; it is more common in those with a **family history of thyroid disease**. Hyperthyroidism has a **large female preponderance**.

The two hormones DEFINITION

The thyroid gland secretes two **iodine-containing** hormones: **thyroxine (T4)** and **tri-iodothyronine (T3)**, produced by the **follicular cells** in response to **thyroid-stimulating hormone (TSH)** from the **anterior pituitary**.

- **T4 is the main hormone secreted** — but it is essentially a **prohormone**.
- **Most of the effects are due to T3**.

The hypothalamic-pituitary-thyroid axis



What thyroid hormones do MECHANISM

- Increase the **basal metabolic rate** — raising *oxygen consumption* and *heat production*.
- Have a role in **growth and development**.
- **Nearly all body tissues are affected**, so an alteration in secretion affects the activity of *all body systems* — hence the wide-ranging symptoms.

Protein binding TECHNICAL NOTE

> **99%** of **both** hormones are protein bound — mainly to **thyroxine-binding globulin (TBG)**.

- Only the **free portion** is active.
- So a change in the amount of TBG alters the measured result without necessarily changing thyroid status.
- This is why **clinical features must always be taken into account** alongside the numbers.

Reading the TFTs

Picture	T4	TSH	Interpretation
Primary thyroid failure	low	high	A fall in serum total or free T4 together with a rise in serum TSH is diagnostic
"Sub-clinical" hypothyroidism	normal	high	Check TFTs yearly — the <i>majority go on to develop clinical hypothyroidism</i>
Hyperthyroidism	high	suppressed	Raised T4 and T3 with suppressed TSH

Never test in isolation WARNING

Because the presenting symptoms are so **varied and non-specific**, biochemical confirmation is necessary — but these tests should **never be used alone** to decide whether treatment is necessary. **Clinical features should always be taken into account.**

- Symptoms of hypothyroidism but **normal TFTs** → thyroxine can be *tried* to see if symptoms improve, as long as TFTs and clinical symptoms are monitored to check for over-treatment (i.e. hyperthyroidism).
- **Abnormal TFTs without any signs or symptoms** → close monitoring, but **no treatment**.

Exam point

- Changes in **thyroid binding globulin (TBG)** would be expected to alter the results — because only the **free** portion of T3 and T4 is active.
- T4 is the main hormone secreted, but T3 does the work: **T4 is essentially a prohormone**.

HYPOTHYROIDISM

The diagnosis DEFINITION

A fall in serum **total or free T4** together with a **rise in serum TSH** is diagnostic of primary thyroid failure. An elevated TSH with a *normal* T4 is termed “**sub-clinical**” hypothyroidism — such patients should have their TFTs checked **yearly**, as the majority go on to develop clinical hypothyroidism.

Causes

HYPOTHYROIDISM

> 90% of cases from the first group

The main causes (> 90%)

- Autoimmune hypothyroidism
- Idiopathic atrophic hypothyroidism
- Post-surgery, radioactive iodine and anti-thyroid drugs

Other causes

- Congenital
- Iodine deficiency

Drugs

- Lithium
- Amiodarone

More than 90% of cases are due to one of the first three.

Clinical features — most common

Onset may be with **non-specific symptoms** and, occasionally, *no symptoms*.

- Tiredness
- Cold intolerance
- Dryness and coarsening of **skin and hair**, subcutaneous swelling
- Hoarseness
- Weight gain
- Hyperlipidaemia

Clinical features — others

- Anaemia
- Depression
- Constipation
- Bradycardia
- Infertility

Thyroxine replacement

START THYROXINE 50 µ DAILY

the usual initial dose; the range is 25–200 µ daily

→ Take at least **HALF AN HOUR BEFORE BREAKFAST** — food stops the drug being absorbed properly

INCREASE BY 50 µ EVERY 3–4 WEEKS

→ Symptomatic improvement is seen within 2–3 weeks

CHECK TSH AFTER 6 WEEKS OF THERAPY

it may be 6 weeks before TSH levels respond fully

→ Adjust the dose by increments of 25–50 µ

ONCE TSH AND T4 ARE NORMAL AND THE PATIENT IS SYMPTOM-FREE

assess adequacy of continuing treatment by **ANNUAL TFTs**

→ Most patients receiving thyroxine require **LIFE-LONG** treatment

Hormone replacement, usually with T4 (thyroxine), is the treatment for hypothyroidism.

Exam point

- The two time constants that matter: **symptoms improve in 2–3 weeks**, but **TSH takes 6 weeks** to respond fully — so do not re-check TSH earlier and do not chase the dose on symptoms alone.
- Thyroxine is absorbed poorly with food: **at least half an hour before breakfast**.

HYPERTHYROIDISM

The diagnosis DEFINITION

Hyperthyroidism — Raised T4 and T3 with suppressed TSH. As only the **free portion** of T3 and T4 is active, changes in the amount of **thyroid binding globulin (TBG)** would be expected to alter the results.

Causes

More than **90%** of cases are due to either:

- **Grave's disease** — an autoimmune disease.
- **Toxic multinodular goitre** — swelling of the thyroid.
- **Other causes:** thyroiditis (due to viral infection); exogenous iodine and iodine-containing drugs e.g. **amiodarone**; excessive T3 and T4 ingestion.

Clinical features

MOST COMMONLY

- Weight loss — but with a normal appetite
- Sweating; heat intolerance
- Increased rate and depth of respiration
- Diarrhoea
- Fatigue, generalised muscle weakness and muscle tremor
- Palpitations; sinus tachycardia or atrial fibrillation; angina; heart failure

+

OTHERS

- Agitation
- Insomnia
- Infertility
- The presenting features in mild cases may MIMIC AN ANXIETY STATE

Three treatment options

- **Anti-thyroid drugs** — e.g. carbimazole
- **Radioactive iodine**
- **Surgery**

Carbimazole MECHANISM

The plasma half-life of carbimazole is relatively **short (6–10 hours)**, but its **biological effect is much longer** — which is what allows a **once daily dose** to be given.

The 'block and replace' regimen

START CARBIMAZOLE 40 mg DAILY
high dose, to 'block' the activity of the thyroid

—> Blocking, plus replacement therapy with thyroxine



MEASURE T4 AFTER 4 WEEKS

—> The decision point



IF NORMAL OR LOW → start thyroxine 100 mcg daily

—> The 'replace' half of the regimen



IF HIGH → repeat at week 6. If STILL high → increase carbimazole to 60 mg daily
when T4 is normal, start thyroxine 100 mcg daily

—> Block harder before replacing



MEASURE T4 4 WEEKS AFTER STARTING THYROXINE
adjust the dose to keep T4 in the normal range

—> Then maintain

Favoured because it reduces the high relapse rate of the previous regimen.

Radioactive iodine (¹³¹I) CLINICAL NOTE

Given as a **tasteless liquid**. It causes necrosis of the overactive thyroid gland without significant local or systemic effects, and with minimal radiation hazard.

- Treatment of choice for **toxic nodular hyperthyroidism** if the goitre is *not large*.
- And for **Grave's disease**, especially on relapse after medical treatment or subtotal thyroidectomy.
- About **two-thirds** are cured by a *single dose*; further doses at intervals of **2–4 months**.

Radioiodine — cautions WARNING

- **Contraindicated during pregnancy.**
- Women who have received treatment should **not get pregnant for 3 months** afterwards.
- All patients need **annual TFTs**: after **20 years**, **50%** will have developed **hypothyroidism**.

Surgery — sub-total thyroidectomy CLINICAL NOTE

It is **rarely first choice**. Beta-blockers — usually **propranolol 20 mg tds** — may provide *temporary symptomatic relief* prior to surgery, or until antithyroid drugs take their effect (about **4 weeks**).